

Chapter 1 Basics concepts of hydrological models: routing

Olivier Bonte

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To get more insight into the routing concepts explained in the theory, a computational guide is given here

1 Reservoir cascade models

1.1 Cascade of linear reservoirs

1.1.1 First order linear reservoir with constant input

A linear reservoir with storage S [m³] and a constant input I [m³/s] is characterised by

$$S = KQ \quad (1)$$

with K [s] the residence time and Q [m³/s] the outflow. The mass balance equation is given by

$$\frac{dS}{dt} = K \frac{dQ}{dt} = I - Q \quad (2)$$

As this is a linear ordinary differential equation, it can be solved analytically using Laplace transforms (see your course in the 2nd bachelor year on [Differential Equations](#)). Remember that although the definition of a Laplace transform might seem complex

$$F(s) = \mathcal{L}\{f(t)\} = \int_0^{\infty} f(t)e^{-st}dt,$$

in practice you only need to apply a limited set of transforms to solve linear ODEs, which are summarised in the table below.

$f(t)$	$F(s) = \mathcal{L}\{f(t)\}$
$H(t)$	$\frac{1}{s}$
t^n	$\frac{n!}{s^{n+1}}$
e^{at}	$\frac{1}{s-a}$
$f'(t)$	$sF(s) - f(0)$

where $H(t)$ is the Heaviside function, which is defined as $H(t) = 0$ for $t < 0$ and $H(t) = 1$ for $t \geq 0$. n is a positive integer and a is a real number. Applying the Laplace transform to equation Equation 2 gives with $Q(t=0) = Q_0 = S_0/K$:

$$K(sQ(s) - Q_0) = \frac{I}{s} - Q(s)$$

Isolating $Q(s)$ gives:

$$Q(s) = \frac{I}{s(1 + Ks)} + \frac{KQ_0}{1 + Ks}$$

We now need to apply the inverse Laplace transform to get $Q(t)$. Trivially, $\mathcal{L}^{-1}(Q(s)) = Q(t)$. For the term containing Q_0 :

$$\mathcal{L}^{-1}\left(\frac{KQ_0}{1 + Ks}\right) = \frac{K}{K}Q_0\mathcal{L}^{-1}\left(\frac{1}{1/K + s}\right) = Q_0e^{-t/K} \quad (3)$$

The term with I is a bit more convolved, but can be solved using partial fraction decomposition:

$$\frac{I}{s(1 + Ks)} = \frac{A}{s} + \frac{B}{1 + Ks}$$

To solve for A , multiply both sides by s and set $s = 0$:

$$\left.\frac{I}{1 + Ks}\right|_{s=0} = A \iff A = I$$

To solve for B , multiply both sides by $(1 + Ks)$ and set $s = -1/K$:

$$\left.\frac{I}{s}\right|_{s=-1/K} = B \iff B = -IK$$

We now need to apply the inverse Laplace transform to get $Q(t)$. Trivially, $\mathcal{L}^{-1}(Q(s)) = Q(t)$. For the term containing Q_0 :

$$\mathcal{L}^{-1}\left(\frac{KQ_0}{1+Ks}\right) = \frac{K}{K}Q_0\mathcal{L}^{-1}\left(\frac{1}{1/K+s}\right) = Q_0e^{-t/K} \quad (4)$$

The term with I is a bit more convolved, but can be solved using partial fraction decomposition:

$$\frac{I}{s(1+Ks)} = \frac{A}{s} + \frac{B}{1+Ks}$$

To solve for A , multiply both sides by s and set $s = 0$:

$$\left.\frac{I}{1+Ks}\right|_{s=0} = A \iff A = I$$

To solve for B , multiply both sides by $(1+Ks)$ and set $s = -1/K$:

$$\left.\frac{I}{s}\right|_{s=-1/K} = B \iff B = -IK$$

Now we can apply the inverse laplace transform on the partial fraction decomposition:

$$\mathcal{L}^{-1}\left(\frac{I}{s} - \frac{IK}{1+Ks}\right) = \mathcal{L}^{-1}\left(\frac{I}{s} - \frac{I}{s+1/K}\right) = I - Ie^{-t/K} = I(1 - e^{-t/K}) \quad (5)$$

Combining Equation 4 and Equation 5 gives the final solution for $Q(t)$:

$$Q(t) = I(1 - e^{-t/K}) + Q_0e^{-t/K} \quad (6)$$

E.g. for $Q_0 = 0.1 \text{ m}^3/\text{h}$ and $K = 7 \text{ h}$, we can plot the outflow $Q(t)$ for different constant inflows I . In the examples that follow, we'll use m^3/h and h for the units instead of the standard SI-units for convenience. Define the analytical solutions as a function:

```
import matplotlib.pyplot as plt
import numpy as np

def Q_analytical_linear(t, I, K, Q0):
    return I * (1 - np.exp(-t / K)) + Q0 * np.exp(-t / K)
```

define the parameters and the three different inflows:

```

Q_0 = 0.1 # m3/h
K = 7 # h
t = np.linspace(0, 40, 200) # h
I = [0.0, 0.06, 0.12] # m3/h

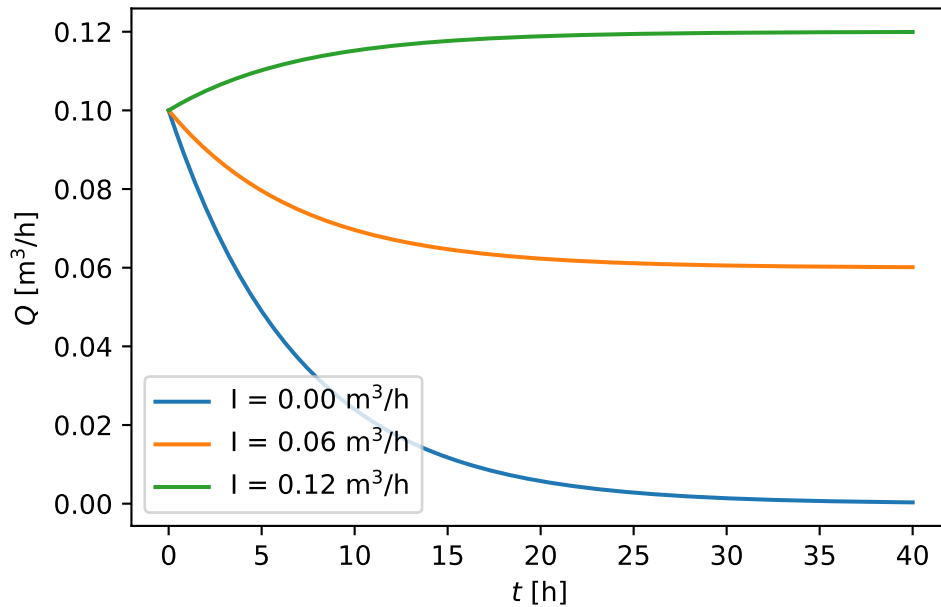
```

to then plot the results:

```

fig, ax = plt.subplots()
for I_ in I:
    Q_ = Q_analytical_linear(t, I_, K, Q_0)
    ax.plot(t, Q_, label=rf"I = {I_:.2f} m3/h")
ax.set_xlabel(r"$t$ [h]")
ax.set_ylabel(r"$Q$ [m3/h]")
ax.legend()

```



Additionally, note that we can automate the algebraic routines above using [SymPy](#). First define all symbols:

```

import sympy as sp

t, K, I, Q0 = sp.symbols("t K I Q0", positive=True)
s = sp.symbols("s")
q, Q = sp.symbols("q, Q", cls=sp.Function)

```

and then define the ODE in the form $K \frac{dQ}{dt} - (I - Q) = 0$:

```
ode_linear = K * sp.diff(Q(t), t) - (I - Q(t))
ode_linear
```

$$-I + K \frac{d}{dt} Q(t) + Q(t)$$

Now take the Laplace transform of the ODE defining $q(s) = \mathcal{L}\{Q(t)\}$:

```
ode_linear_laplace = sp.laplace_transform(ode_linear, t, s, noconds=True)
ode_linear_laplace = sp.laplace_correspondence(ode_linear_laplace, {Q: q})
ode_linear_laplace
```

$$-\frac{I}{s} + K(sq(s) - Q(0)) + q(s)$$

We now insert the initial condition $Q(0) = Q_0$:

```
ode_linear_laplace_ic = sp.laplace_initial_conds(ode_linear_laplace, t, {Q: [Q0]})
ode_linear_laplace_ic
```

$$-\frac{I}{s} + K(-Q_0 + sq(s)) + q(s)$$

In the equation above, the only unknown is $q(s)$, which can be solved for:

```
q_s_solved = sp.solve(ode_linear_laplace_ic, q(s))[0]
q_s_solved
```

$$\frac{I + KQ_0s}{s(Ks + 1)}$$

Now apply the inverse Laplace transform to get $Q(t)$:

```
Q_solved = sp.simplify(sp.inverse_laplace_transform(q_s_solved, s, t))
Q_solved
```

$$\left(Ie^{\frac{t}{K}} - I + Q_0\right)e^{-\frac{t}{K}}$$

Which is identical to the analytical solution Equation 6 derived above.

1.1.2 Second order linear reservoir with constant input

We now get two reservoirs in series:

$$\begin{aligned}K^{(1)} \frac{dQ^{(1)}}{dt} &= I - Q^{(1)} \\K^{(2)} \frac{dQ^{(2)}}{dt} &= Q^{(1)} - Q^{(2)}\end{aligned}$$

The same SymPy approach can be extended to solve this coupled system. To simplify the solution, we'll assume:

- The second reservoir is initially empty, i.e. $Q^{(2)}(0) = 0$.
- The first reservoir has an initial outflow $Q^{(1)}(0) = Q_0$ as it is assumed non-empty.
- The first and second reservoirs have the same residence time, i.e. $K^{(1)} = K^{(2)} = K$.

We define two unknown functions and their Laplace transforms:

```
q1, q2, Q1, Q2 = sp.symbols("q1 q2 Q1 Q2", cls=sp.Function)
```

Write both ODEs in the form $(\dots) = 0$:

```
ode1 = K * sp.diff(Q1(t), t) - (I - Q1(t))
ode2 = K * sp.diff(Q2(t), t) - (Q1(t) - Q2(t))
display(ode1, ode2)
```

$$-I + K \frac{d}{dt} Q_1(t) + Q_1(t)$$

$$K \frac{d}{dt} Q_2(t) - Q_1(t) + Q_2(t)$$

Take the Laplace transform using $q_1(s) = \mathcal{L}\{Q_1(t)\}$ and $q_2(s) = \mathcal{L}\{Q_2(t)\}$:

```
ode1_laplace = sp.laplace_transform(ode1, t, s, noconds=True)
ode1_laplace = sp.laplace_correspondence(ode1_laplace, {Q1: q1})

ode2_laplace = sp.laplace_transform(ode2, t, s, noconds=True)
ode2_laplace = sp.laplace_correspondence(ode2_laplace, {Q1: q1, Q2: q2})
```

Insert the initial conditions:

```
ode1_laplace_ic = sp.laplace_initial_conds(ode1_laplace, t, {Q1: [Q0], Q2: [0]})
ode2_laplace_ic = sp.laplace_initial_conds(ode2_laplace, t, {Q1: [Q0], Q2: [0]})
display(ode1_laplace_ic, ode2_laplace_ic)
```

$$-\frac{I}{s} + K(-Q_0 + sq_1(s)) + q_1(s)$$

$$Ksq_2(s) - q_1(s) + q_2(s)$$

Solve the algebraic system for $q_1(s)$ and $q_2(s)$:

```
q_s_solved = sp.solve([ode1_laplace_ic, ode2_laplace_ic], [q1(s), q2(s)], dict=True)[0]
q2_s = q_s_solved[q2(s)]
q2_s
```

$$\frac{I + KQ_0s}{K^2s^3 + 2Ks^2 + s}$$

To get a solution closer to the notation of the theory, we'll first break up into partial fractions:

```
q2_s_partial = sp.apart(q2_s, s)
q2_s_partial
```

$$-\frac{IK}{Ks+1} + \frac{I}{s} - \frac{K(I-Q_0)}{(Ks+1)^2}$$

Finally, apply the inverse Laplace transform to obtain $Q_2(t)$.

```
Q2_solved = sp.inverse_laplace_transform(q2_s_partial, s, t)
Q2_solved
```

$$I - Ie^{-\frac{t}{K}} - \frac{t(I-Q_0)e^{-\frac{t}{K}}}{K}$$

Given that the second reservoir is initially empty and receives no external input I , Q_2 is entirely defined by Q_1 and $K^{(1)} = K$. So if you solve for Q_2 (in the s -domain $q_2(s)$) with an assumed known $Q_1(t)$ function (in the s -domain $q_1(s)$), you get:

```
q2_s_bis = sp.solve(ode2_laplace_ic, q2(s))[0]
q2_s_bis
```

$$\frac{q_1(s)}{Ks+1}$$

This concept can be generalised in what is called a **transfer function**, which is defined as the ratio of the Laplace transforms of the output and input of a system. In this case, the transfer function $H(s)$ is defined as:

$$H(s) = \frac{q_2(s)}{q_1(s)} = \frac{1}{1 + Ks}$$